Data Analysis

Course Overview

Data Analysis with Python
Burkina Institute of Technology - L3 Computer Science
Semester 5 - ECU 2SCD1502 (TU Data Science, SCD1502)

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Welcome

This course is about one thing: turning raw data into answers with Python.

You already know how to write Python and you have seen basic statistics. We are now going to put both together on real datasets: grades, sales, weather, survey results.

By the end of the semester, you will be able to load a messy file, clean it, describe it, plot it and defend a conclusion in front of someone who did not do the work.

What you need before we start

Prerequisites from the syllabus

- Python programming: variables, lists, dictionaries, functions, loops
- Statistics: mean, median, standard deviation, basic probability

What we will check in the first session

If one of these points is unclear for you, tell me during the first practical. We will fix it before it becomes a problem.

Course plan

Module	Title
1	Introduction to Data Analysis with Pandas
2	Graphical Data Representation with Matplotlib
3	Acquiring External Data: CSV, JSON, XML, web
4	Statistical Analysis with NumPy and SciPy

Module 1 - Pandas and DataFrames

What we will see

- The DataFrame as the central structure of data analysis
- Selecting, filtering and indexing rows and columns
- Missing data, merging tables, working with dates
- Statistics on columns, scaling, building your own metrics

This is the longest module. Everything else depends on it.

Module 2 - Data visualisation

What we will see

- Why a chart is often more convincing than a table
- The Matplotlib figure: axes, titles, labels, legends
- Line, bar, histogram and scatter plots on real problems
- Reading a chart correctly and avoiding misleading graphics

Module 3 - External data

What we will see

- Structured formats: CSV, JSON and XML
- Reading and writing files from Python and from Pandas
- Getting data from a URL or a public API
- The errors you will meet: wrong separator, wrong encoding, bad JSON

Module 4 - Statistical analysis

What we will see

- NumPy arrays and matrix calculation
- SciPy: quantiles, distributions, statistical laws
- Comparing populations and measuring association
- Automatic classification with k-means, performance basics

Tools we will use

Software

- Python 3.11 or newer
- Jupyter Notebook or JupyterLab
- VS Code (optional)
- Git for the project (optional)

Libraries (versions used in class)

- pandas 2.2
- NumPy 2.2
- Matplotlib 3.10
- SciPy 1.15

If your version differs slightly, most examples still work. I will tell you when a version matters.

Installing the environment

To run once on your machine

```
python -m venv .venv
.venv\Scripts\activate          (Windows)
source .venv/bin/activate       (Linux / macOS)
pip install pandas numpy matplotlib scipy jupyter
```

Check

```
python -c "import pandas; print(pandas.__version__)"
2.2.3
```

If the import fails

```
>>> import pandas
ModuleNotFoundError: No module named 'pandas'
```

Cause: the library was installed in another Python, or the virtual environment is not activated.

Fix: activate the environment, then run `pip install pandas` again. Check with `pip list` that pandas appears.

How a session works

1. Short explanation of a notion (10 to 20 minutes)
2. Live demonstration in a notebook, you type along
3. Exercise you do alone or in pairs
4. Correction together, with the errors we met

Bring your laptop to every session. Data analysis is not learned by watching.

Assessment

From the syllabus

Component	Weight
Continuous assessment	01
Project / Practical work	01
Final exam	00

There is no final written exam. Your grade comes from practical work during the semeste.

What I expect in your work

- Code that runs from top to bottom without manual steps
- Comments that explain why, not what
- Every chart has a title, axis labels and units
- Every conclusion is backed by a number or a figure

A notebook that produces a wrong answer with a clear method gets more credit than a right answer nobody can reproduce.

Bibliography

1. Tarek Ziadé, *Python programming: Design and optimization*, Eyrolles, 2009
2. Alberto Boschetti, Luca Massaron, *Python Data Science Essentials*, 3rd ed., Packt Publishing, 2018

These are the two references of the official syllabus. The second one is the closest to what we do in class.

Online documentation

Bookmark these four. They are the reference when my slides and your memory disagree.

- pandas: https://pandas.pydata.org/docs/
- NumPy: https://numpy.org/doc/stable/
- Matplotlib: https://matplotlib.org/stable/
- SciPy: https://docs.scipy.org/doc/scipy/

Contact

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Ask questions in class first. If you write to me, put the course name and your group in the subject line and attach the notebook that causes the problem.

Questions?

What kind of data would you like to analyse this semester? Think about it, we will use your ideas for the project.